

Research Requirement Document

Paper Title: Multimodal Satellite Change Detection for Urban Development and Environmental Monitoring

Course / Track: On-the-Job Training (OJT) / Data Science - Research

Track section filled: Artificial Intelligence / Machine Learning / Remote Sensing / Computer Vision

1. Document Overview & Metadata

Field	Description / Student Inputs
Project Title	Multimodal Satellite Change Detection for Urban Development and Environmental Monitoring
Project Track	Data Science - Research
Team Members & IDs	Manish Y M
Assigned Mentor	Soumen Mukherjee
Primary deliverable	Research paper and experimentally evaluated multimodal satellite change detection framework
Domain	Remote Sensing, Multimodal Artificial Intelligence, Computer Vision, Deep Learning and Geospatial Intelligence
Grounding paper	Hafner et al., Sentinel-1 and Sentinel-2 Data Fusion for Urban Change Detection using a Dual Stream U-Net; Song et al., A Dual-Modal Framework Integrating SAR-Based Change Screening and Optical-Scene-Informed Identification for High-Frequency Monitoring of Construction-Ready Bare Land
Dataset	Primary benchmark: OSCD enriched with Sentinel-1 SAR. Any additional evaluation data will be finalized after verifying data availability and annotation feasibility.

2. Research Problem & Motivation

2.1 What real-world or scientific problem does this work address?

Rapid urbanization and environmental change require frequent and reliable monitoring of Earth's surface. Satellite remote sensing provides large-scale observations, but optical and SAR sensors have different strengths. Optical imagery provides rich spectral and semantic information but may be unavailable because of clouds, while SAR provides all-weather observations but can be difficult to interpret semantically. The research therefore addresses robust multimodal satellite change detection when synchronous cloud-free optical imagery is unavailable.

2.2 What exactly is going wrong or missing?

- Many multimodal change detection approaches assume temporally synchronized SAR and optical observations.
- SAR observations can produce false alarms because agricultural, vegetation, soil and other environmental changes may resemble meaningful urban changes.
- Historical optical imagery can provide semantic context, but its usefulness may decrease as its temporal distance from the target SAR observation increases.
- Small or low-intensity changes may be missed by SAR-based candidate screening.
- Existing asynchronous work is focused on construction-ready bare land and does not explicitly cover broader transitions such as urban renewal and demolition.
- Richer SAR information such as multi-polarization features and InSAR coherence remains an open direction.

2.3 Who or what is affected by this problem?

- Researchers working on remote sensing, multimodal fusion, computer vision and satellite change detection.
- Urban planners monitoring urban expansion, construction and land conversion.
- Government and infrastructure agencies requiring frequent large-area monitoring.
- Environmental monitoring organizations studying land-cover and surface changes.
- Society, through the need for more reliable and scalable urban and environmental monitoring.

2.4 Why is this research important? Explain its importance to:

Researchers – The study investigates multimodal change detection under asynchronous acquisition conditions and examines how temporal separation affects the usefulness of historical optical semantic information.

Practitioners/Industry – The approach can reduce dependence on obtaining synchronous cloud-free optical imagery and can support more frequent monitoring using current SAR observations together with available historical optical imagery.

Users/Society – More reliable change maps can support urban planning, land-use management, infrastructure monitoring and environmental observation.

3. Research Gap

Existing research has demonstrated the value of SAR and optical fusion for urban change detection. Hafner et al. proposed a Dual Stream U-Net that processes SAR and optical data separately before decision-level fusion, improving over their unimodal and feature-level fusion baselines. However, that setup uses paired bitemporal SAR and optical observations. Song et al. later demonstrated that historical optical imagery can provide semantic guidance without requiring synchronous T2 optical imagery, reducing false alarms in monitoring construction-ready bare land.

The remaining gap is the systematic investigation of robust asynchronous multimodal change detection across temporal gaps, richer SAR configurations and broader urban and environmental pseudo-change conditions. In particular, the effect of historical optical temporal distance on precision, false alarms and spatial agreement requires further investigation in a broader change detection setting.

4. Area of Research

Questions the research tries to answer:

- RQ1: Can historical Sentinel-2 optical information improve Sentinel-1 SAR-based change detection when synchronous post-change optical imagery is unavailable?
- RQ2: How does the temporal gap between historical optical imagery and the target SAR observation affect change detection performance?
- RQ3: Can additional SAR information, particularly VV and VH polarization, improve discrimination between meaningful urban development and environmental or agricultural pseudo-changes?
- RQ4: How does the proposed asynchronous multimodal approach compare with SAR-only, optical-only, conventional fusion and dual-stream approaches?
- RQ5: Under what temporal and environmental conditions does historical optical information provide the greatest benefit?

5. Research Objectives

What do you want to achieve?

1. To develop an asynchronous multimodal satellite change detection framework integrating Sentinel-1 SAR and historical Sentinel-2 optical imagery.
2. To investigate the effect of temporal separation between historical optical imagery and the target SAR observation.
3. To investigate whether additional SAR polarization information can improve change discrimination.
4. To reduce false positives associated with agricultural, seasonal and environmental pseudo-changes.

5. To evaluate the proposed framework using Precision, Recall, F1-score, IoU and False Alarm Rate.
6. To compare the proposed approach against single-modality, feature-level fusion and dual-stream baselines.
7. To analyze the limitations of historical optical semantic guidance under increasing temporal gaps.

6. Related Work

Author & Year	Existing Method	Main Contribution	Limitation
Daudt et al., 2018	Fully Convolutional Siamese Networks	Established Siamese architectures for bitemporal change detection.	Primarily developed around optical imagery
Hafner et al., 2022	Dual Stream U-Net	Processes SAR and optical data in separate streams and combines change information at a later stage.	Uses paired bitemporal observations and remains challenged by small urban changes.
Ebel et al., 2021	Multimodal Siamese Fusion	Investigates fusion of multimodal features for supervised change detection.	Multimodal improvement and modality interaction remain challenging.
Song et al., 2026	SCS-OI with ASG-Net	Uses SAR for high-recall candidate screening and historical optical imagery for semantic refinement without synchronous T2 optical imagery.	Focused on construction-ready bare land; small changes, broader transitions and richer SAR representations remain open.
Saha et al., 2022	Self-Supervised Multisensor Change Detection	Investigates label-efficient multisensor change detection.	Further investigation is needed for the proposed asynchronous urban and environmental setting.
Jiang et al., 2022	DSNUNet	Combines Sentinel-1 and Sentinel-2 information for forest	Application-specific and focused on forest change.

		change detection.	
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Overall gap: Existing research has successfully addressed multimodal SAR-optical fusion, modality-specific processing and asynchronous historical-optical guidance, but robust evaluation across temporal gaps, richer SAR representations and environmental pseudo-change conditions remains insufficiently addressed.

7. Proposed Approach

The proposed research will develop a multimodal satellite change detection framework that combines the complementary strengths of Sentinel-1 SAR and Sentinel-2 optical imagery for urban development and environmental monitoring.

The central idea is to use bitemporal SAR observations to identify physical surface changes while using historical optical imagery to provide semantic and land-cover context, even when a synchronous optical observation is unavailable. The research will specifically investigate how the temporal gap between the historical optical image and the target SAR observation affects change detection performance.

The proposed approach will consist of five main stages:

Data / Input → Preprocessing → Multimodal Model → Optimization → Change Map

The framework will be evaluated against single-modality and existing multimodal approaches to determine whether asynchronous multimodal information can improve detection accuracy, reduce false alarms and provide more reliable monitoring of urban and environmental changes.

7.1 Data / Input

- Sentinel-1 SAR imagery will provide all-weather structural information and bitemporal observations for change representation.
- Sentinel-2 optical imagery will provide historical multispectral semantic context.
- The primary benchmark will be the OSCD dataset enriched with Sentinel-1 SAR data. The original OSCD benchmark contains bitemporal Sentinel-2 imagery and pixel-level urban change labels, while the Hafner study enriched the dataset with Sentinel-1 SAR observations.
- Historical optical observations will be selected before the target SAR observation to create controlled asynchronous temporal gaps.

7.2 Preprocessing

- SAR: thermal noise removal, radiometric calibration, terrain correction, log scaling to decibels, noise masking, orbit consistency and normalization.
- SAR: VV and VH will be considered where consistently available. The Hafner study used VV because dual polarization was not available for all early Sentinel-1 acquisitions.
- Optical: cloud and quality masking, band selection, resampling, spatial registration and normalization.

- Spatial alignment will be applied so that SAR, optical imagery and labels correspond to the same geographic regions.
- Training data may use random crops, change-aware sampling and geometric augmentation such as flips and 90 degree rotations.

7.3 Proposed Method / Model

- The proposed framework will use modality-specific processing so that SAR and optical information can retain their distinct characteristics. A SAR branch will encode bitemporal SAR observations and represent physical surface change. A historical optical branch will encode the pre-change semantic context. The resulting representations will be fused and passed to a decoder to generate a pixel-level change probability map.
- The study may evaluate asymmetric cross-modal attention as one fusion mechanism. However, asymmetric cross-modal attention itself will not be claimed as the main novelty because Song et al. already introduced ASG-Net using SAR features as Query and historical optical features as Key and Value.
- The research emphasis will instead be on asynchronous robustness, temporal-gap behavior, richer SAR information and false-alarm analysis.

7.4 Adaptation / Optimization

If applicable:

- Supervised training using pixel-level change labels.
- Controlled training settings across all baseline and proposed configurations.
- Optimization and loss functions will be selected through validation experiments rather than assumed as a research contribution.
- Multiple random seeds will be used to reduce the influence of training randomness.
- Early stopping, learning-rate scheduling and class-imbalance handling may be applied where appropriate.

7.5 Output

- Pixel-level binary change map.
- Change probability map.
- Precision, Recall, F1-score and IoU measurements.
- False-alarm analysis and qualitative error maps.
- Temporal-gap performance analysis.

8. Key Innovation / Novelty

The key novelty of this research is an experimentally evaluated framework for robust asynchronous SAR-optical change detection that investigates how historical optical semantic information and richer SAR observations affect change detection under varying temporal gaps and environmental pseudo-change conditions.

- Systematic asynchronous multimodal evaluation.
- Controlled temporal-gap analysis of historical optical information.
- Evaluation of richer SAR information, particularly VV and VH polarization where available.

- Explicit false-alarm and pseudo-change analysis.
- Evaluation beyond a single construction-ready bare-land scenario where suitable data are available.
- Systematic comparison of unimodal, conventional multimodal, dual-stream and asynchronous configurations.

9. Experimental Design

The study will be experimental, empirical, computational, comparative, system-building and exploratory.

- Dataset: OSCD enriched with Sentinel-1 SAR as the primary benchmark.
- Experimental environment: Python, PyTorch, Google Earth Engine and standard geospatial raster-processing tools.
- Train/validation/test strategy: preserve the official OSCD train/test separation for benchmark comparisons. Custom data, if used, will use geographically separated training, validation and test regions where feasible.
- Baseline methods: SAR-only, optical-only, feature-level SAR-optical fusion and Dual Stream U-Net.
- Experimental scenarios: modality comparison, fusion comparison, temporal-gap analysis, SAR polarization ablation, pseudo-change analysis and small-change analysis where labels permit.
- Repeated experiments: multiple independent runs with different random seeds will be used for the main deep-learning comparisons.
- Controlled variables: preprocessing, data split, training budget, augmentation, evaluation metrics and thresholding procedure will be kept consistent across comparable experiments.

10. Baseline / Comparative Models

What will you compare your method against?

Method	Purpose
SAR-only U-Net	Single-modality structural baseline.
Optical-only U-Net	Single-modality semantic baseline.
Feature-level SAR + Optical U-Net	Conventional multimodal fusion baseline.
Dual Stream U-Net	Strong multimodal baseline using separate SAR and optical streams.
Asynchronous SAR + historical optical	Reference for asynchronous semantic

configuration	guidance.
Proposed framework	Final configuration evaluated under the proposed research conditions.

11. Evaluation Metrics

- Precision = $TP / (TP + FP)$. Measures the reliability of positive change predictions.
- Recall = $TP / (TP + FN)$. Measures the completeness of detected changes.
- F1-score = $2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$. Provides a balanced measure of Precision and Recall.
- IoU = $TP / (TP + FP + FN)$. Measures spatial overlap between prediction and reference.
- False Alarm Rate = $FP / (FP + TN)$. Particularly important for evaluating environmental and agricultural pseudo-changes.
- Standard deviation across repeated runs will be reported where applicable.
- Inference time and computational cost may be reported as secondary system metrics.

12. Ablation Study

Determine which parts of your proposed method actually help.

Configuration	Performance and Purpose
SAR only	Establish the structural single-modality baseline.
Optical only	Establish the semantic single-modality baseline.
SAR + Optical feature fusion	Measure conventional multimodal fusion.
Dual Stream	Measure modality-specific processing and later fusion.
Historical optical removed	Measure the contribution of historical semantic information.
VV only	Measure single-polarization SAR performance.
VH only	Measure cross-polarization SAR performance.
VV + VH	Measure the benefit of multi-polarization information.

Different temporal gaps	Measure the robustness of historical optical guidance as temporal separation increases.
Full proposed configuration	Evaluate the combined contribution of the selected components.

13. Implications and Impact

- The study can provide evidence about how asynchronous SAR-optical fusion behaves under realistic satellite acquisition constraints.
- Temporal-gap results can provide practical guidance on how recent historical optical imagery should be for reliable semantic guidance.
- False-alarm analysis can clarify the effect of agricultural and environmental pseudo-changes on satellite change detection.
- Urban monitoring applications may include urban expansion, construction, land conversion and redevelopment monitoring.
- Environmental applications may include land-cover disturbance and agricultural or seasonal change analysis.
- The framework may support monitoring systems that are less dependent on synchronous cloud-free optical observations.

14. Limitations

- The primary benchmark contains a limited number of geographic sites and may not represent all urban and environmental conditions.
- Historical optical information may become less representative as temporal distance increases.
- SAR observations are affected by speckle, viewing geometry, terrain, soil moisture and scattering characteristics.
- Small or low-intensity changes may remain difficult to detect.
- Reliable labels for environmental pseudo-changes and broader urban transitions may be limited.
- Multimodal deep-learning models can require greater computational resources than single-modality models.
- Generalization to cities, climates and geographic regions outside the evaluated data remains uncertain.

15. Threats to Validity

Common categories:

- **Internal validity** – Potential threats include data leakage, inconsistent preprocessing, different hyperparameter budgets, random initialization, class

imbalance and threshold selection. These will be mitigated through fixed data splits, standardized preprocessing, controlled training and repeated runs.

- **External validity** – Evaluation on a limited number of geographic regions may not guarantee global generalization. Geographic separation and additional regions will be considered where suitable data are available.
- **Construct validity** – A pixel-level change label identifies surface change but may not always identify the semantic cause of that change. Error analysis will therefore distinguish urban development from environmental or agricultural pseudo-changes where the data support such analysis.
- **Reproducibility** – The study will document data sources, acquisition dates, preprocessing, model configuration, hyperparameters, training seeds and evaluation procedures. Code and experiment configurations will be shared where licensing permits.

16. Expected Contribution

Clearly state what the research contributes.

This research is expected to contribute:

1. **Methodological contribution:** A framework for asynchronous multimodal change detection using current SAR observations and historical optical semantic information.
2. **Technical contribution:** An experimentally evaluated multimodal pipeline incorporating bitemporal SAR, historical optical context and multi-polarization SAR where available.
3. **Experimental contribution:** A systematic study of modality contribution, fusion strategy, temporal gap, SAR polarization, pseudo-changes and small-change behavior.
4. **Practical contribution:** A monitoring approach that reduces dependence on synchronous cloud-free optical observations.

17. Conclusion & Future Work (Brief)

Short Conclusion

The proposed research investigates robust asynchronous multimodal satellite change detection for urban development and environmental monitoring. It builds on established SAR-optical fusion and historical optical semantic guidance while focusing on unresolved questions involving temporal asynchrony, richer SAR information and false-alarm behavior. The central experimental objective is to determine how historical optical information changes in usefulness as its temporal distance from the target SAR observation increases.

Future Work :

- Larger and more geographically diverse datasets.
- More diverse urban and environmental transition types.
- InSAR coherence and additional SAR-derived features.

- Longer satellite time series and temporal deep-learning models.
- Cross-city and cross-region generalization.
- Label-efficient and self-supervised multimodal learning.
- Urban demolition and renewal detection.
- Near real-time or large-scale operational deployment.

Source Basis

This PRD is grounded primarily in the two research papers supplied for the project. Hafner et al. establish the Dual Stream U-Net approach for Sentinel-1 and Sentinel-2 urban change detection and describe the OSCD dataset enrichment and experimental setup. Song et al. establish the SCS-OI framework, historical optical semantic guidance, temporal-gap experiments and limitations involving small changes, local environmental conditions, broader transitions, multi-polarization SAR and time-series modeling.

Hafner et al.: Sentinel-1 and Sentinel-2 Data Fusion for Urban Change Detection using a Dual Stream U-Net.

Song et al.: A Dual-Modal Framework Integrating SAR-Based Change Screening and Optical-Scene-Informed Identification for High-Frequency Monitoring of Construction-Ready Bare Land.